Ain Shams University – Faculty of Engineering I-CHEP

CSE356: Internet of Things

Part 2: Simulation Report

Smart Mosque Management System

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Contents

1. Introduction

The Smart Mosque Management System is a simulated IoT-based environment developed in Cisco Packet Tracer to improve operational efficiency, worshipper comfort, energy conservation, and water management inside mosques.

The system follows a Sense → Process → Act → Monitor operational model where sensors continuously monitor environmental conditions, the IoT controller processes sensor data, automation conditions trigger actuator responses, and administrators supervise the system through a centralized dashboard.

The project focuses on:

- Smart energy management
- Occupancy-aware automation
- Environmental comfort control
- Water conservation
- Real-time monitoring

The implementation demonstrates how IoT technologies can support sustainable public facilities while reducing operational waste and improving responsiveness during high-attendance periods such as Friday prayers and Ramadan.

2. System Topology and Infrastructure

The simulation environment is divided into three functional zones:

- Prayer Hall
- Wudu Area
- Exterior / Entrance Area

The infrastructure is built around a centralized IoT architecture where sensors, actuators, servers, and wireless access points communicate through a local network.

2.1 Core Infrastructure Components

The system includes:

- Cisco 2960-24TT switch
- IoT Registration Server
- DHCP Server
- DNS Server
- HTTP/HTTPS Dashboard Server
- Wireless Access Points

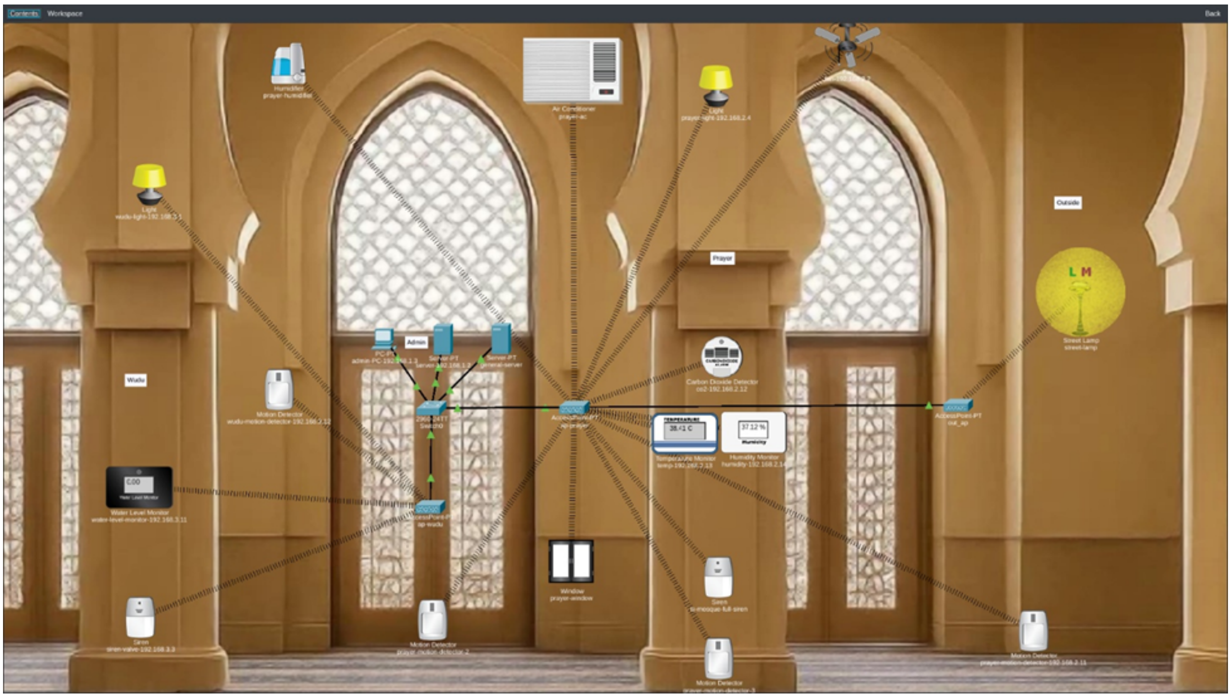


Figure 1: Overall smart mosque topology

2.2 IoT Device Registration

All IoT devices are registered through the IoT Registration Server to allow centralized automation and monitoring.

Figure 2: IoT Registration Server device inventory

3. Network Services and Security

The system uses multiple network services to maintain secure and reliable communication.

3.1 DHCP and DNS Services

The DHCP server dynamically assigns IP addresses to connected IoT devices, while the DNS server resolves dashboard and server names locally.

Figure 3: DHCP configuration

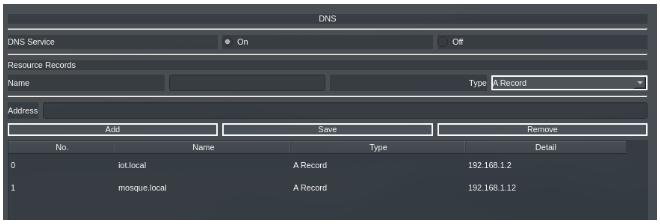


Figure 4: DNS configuration

3.2 Wireless Security

Wireless communication is protected using WPA2-PSK authentication to prevent unauthorized device access.

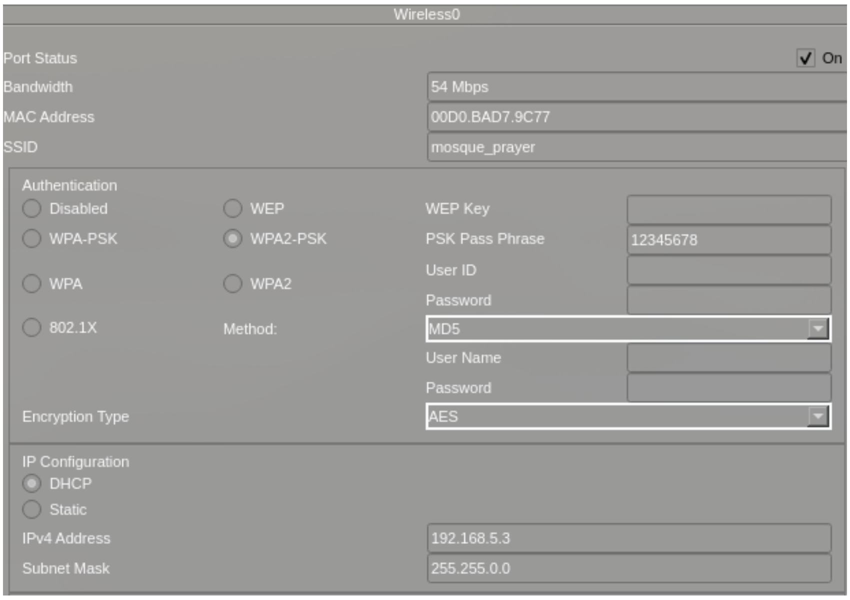


Figure 5: Wireless security configuration

3.3 Connectivity Verification

The simulation verifies successful network communication between sensors and the IoT controller.

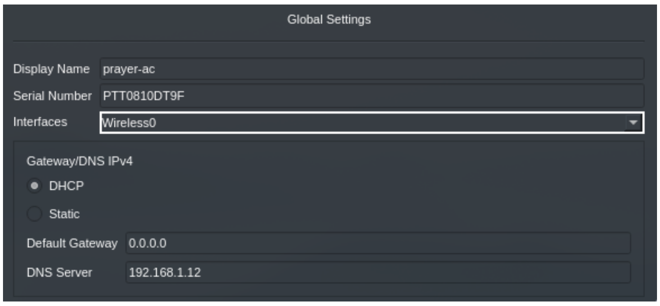


Figure 6: Sensor IP assignment

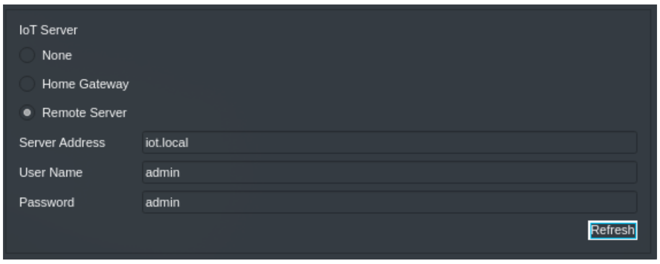


Figure 7: Sensor-to-server communication

4. Implemented Smart Features

4.1 Occupancy-Based Lighting

Motion sensors automatically activate lighting when occupancy is detected inside the mosque.

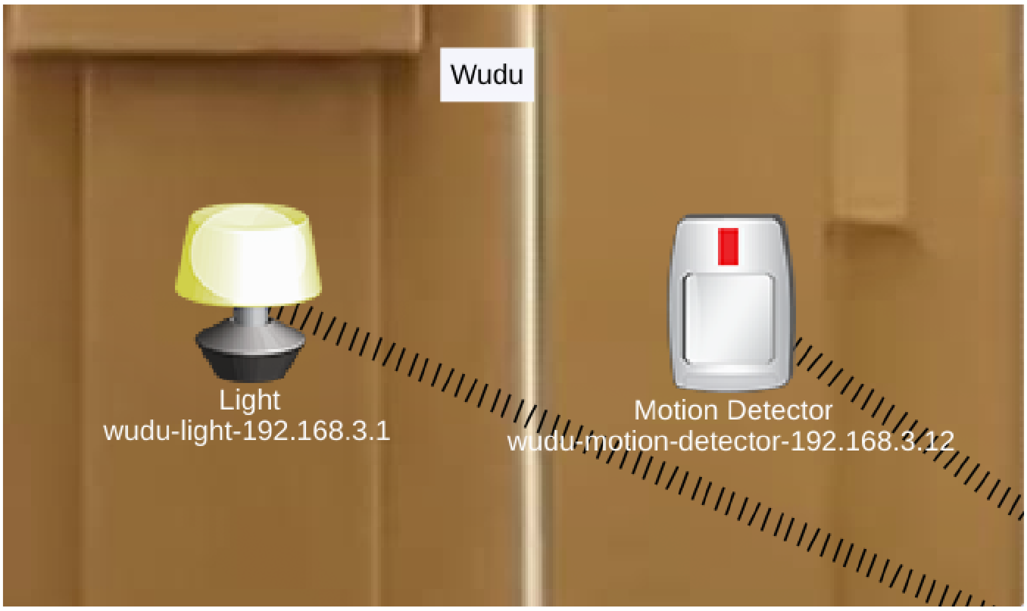


Figure 8: Lighting activation after motion detection

4.2 Environmental Control

The system monitors:

- Temperature
- Humidity
- CO_2

Environmental automation thresholds are shown below.

Parameter	Threshold	Action
Temperature	> 28°C	Activate fan
Temperature	> 32°C	Activate AC
Humidity	< 30%	Activate humidifier
Humidity	≥ 60%	Open smart window
CO_2	High threshold exceeded	Activate ventilation

Table 1: Environmental automation thresholds

4.3 HVAC Verification

Environmental conditions were manipulated through the Packet Tracer environment controls to test HVAC activation behavior.

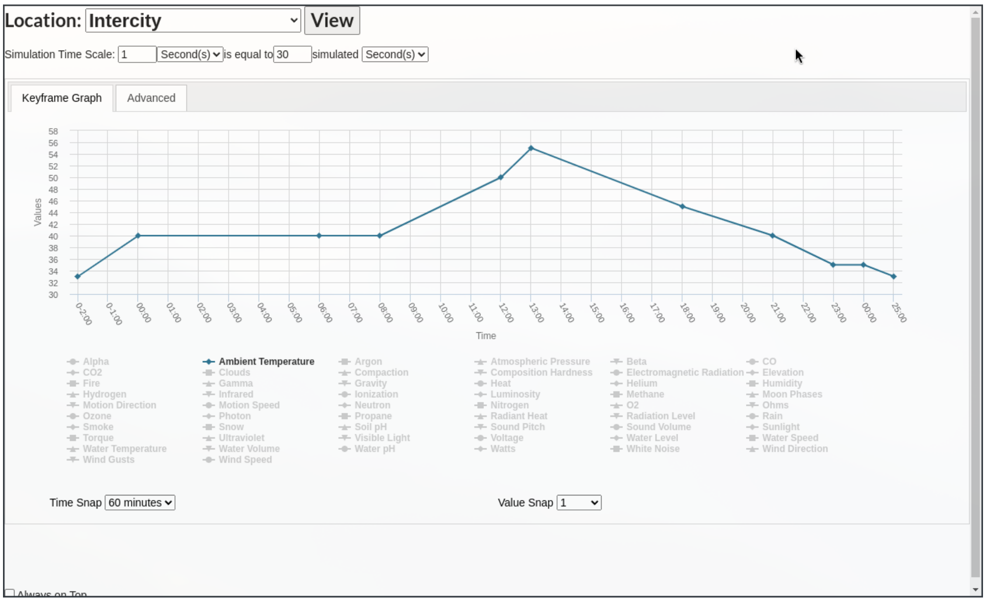


Figure 9: Temperature manipulation testing

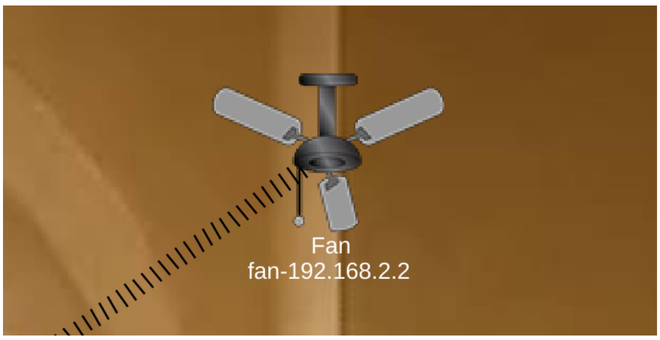


Figure 10: Normal state

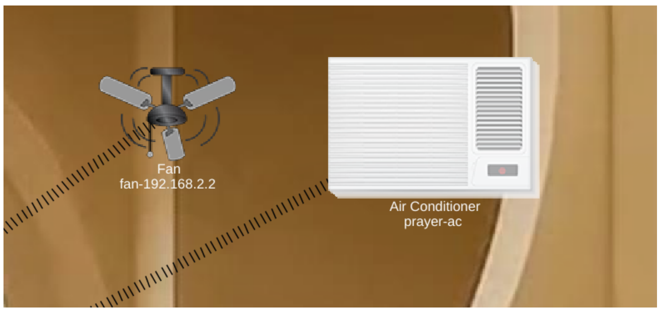


Figure 11: AC and fan activation

4.4 Humidity Comfort Control

When humidity drops below 30%, the humidifier activates automatically to maintain indoor comfort.

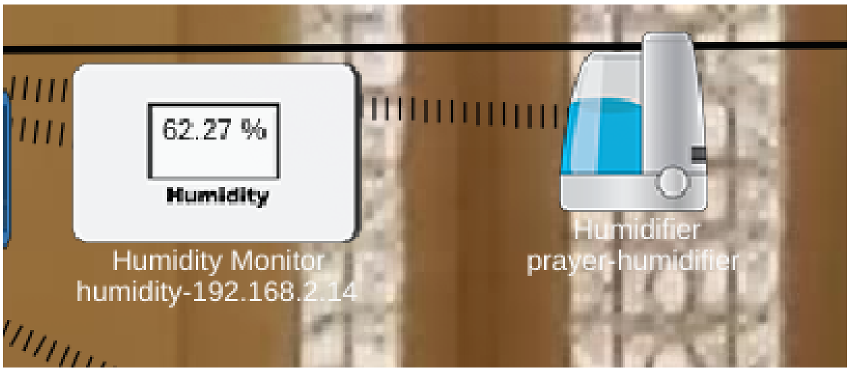


Figure 12: Humidifier activation

4.5 Smart Window Automation

The smart window supports passive ventilation:

- Opens when humidity is high and AC is OFF
- Closes automatically when AC activates

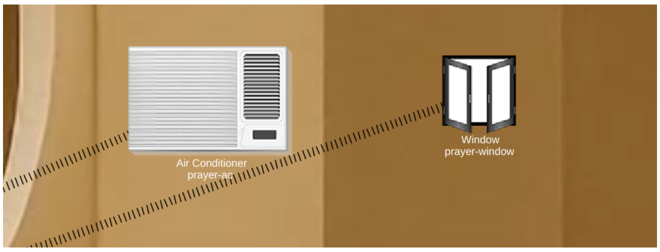


Figure 13: Window open

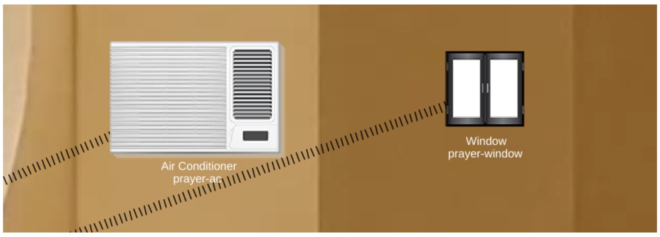


Figure 14: Window closed

4.6 Water Safety Monitoring

Water flow is monitored continuously to detect leaks or taps left open.

Due to Cisco Packet Tracer limitations, a siren is used to simulate solenoid valve activation.

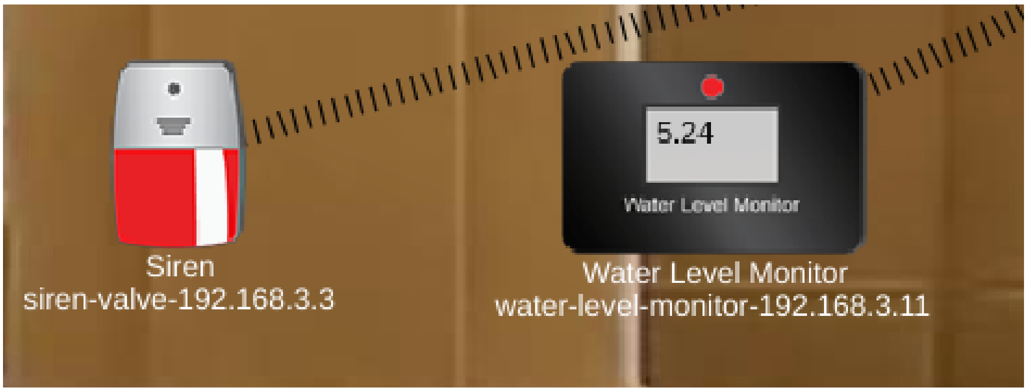


Figure 15: Water leak alert activation

4.7 Occupancy Full Detection

When all prayer hall motion sensors activate simultaneously, the occupancy-full siren is triggered automatically.

4.8 Street Lamp Automation

Exterior lighting activates automatically depending on environmental conditions and occupancy behavior.

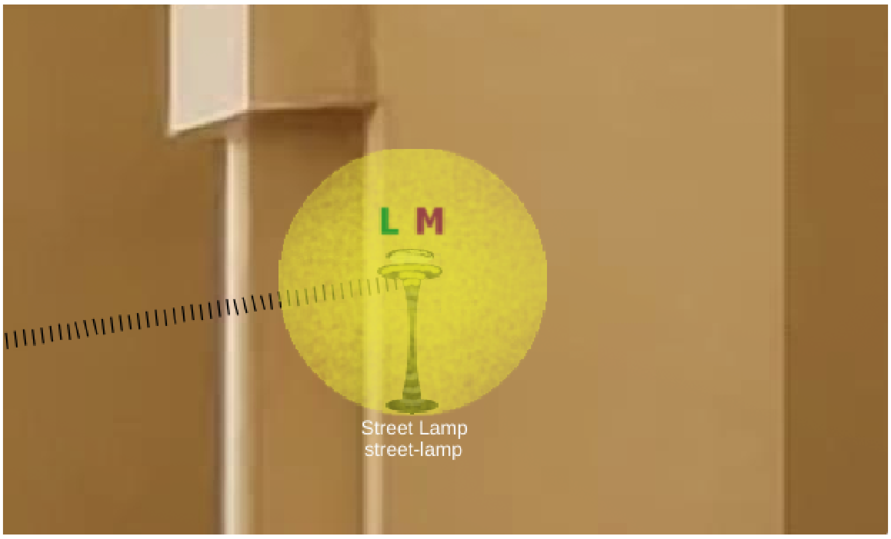


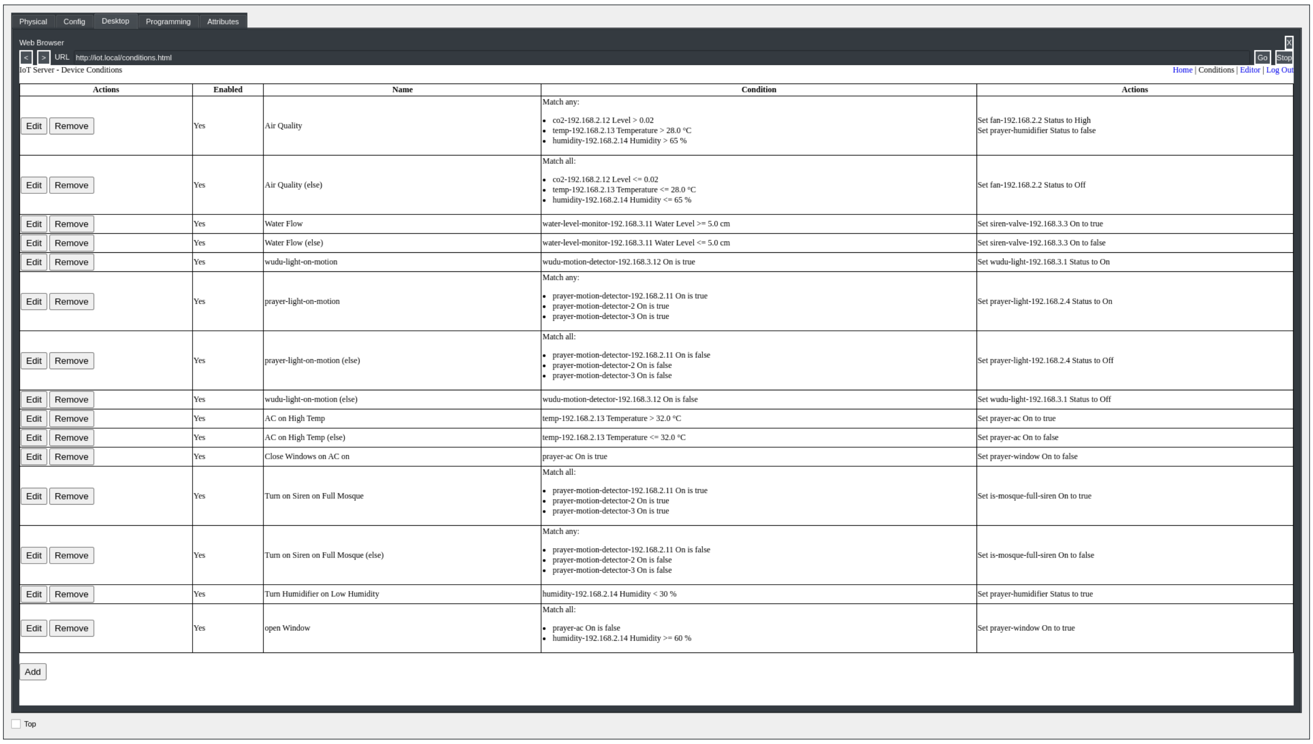
Figure 16: Street lamp automation

5. Automation Logic

The system behavior is controlled through IoT Server automation conditions.

The implemented logic includes:

- Occupancy-based lighting
- Temperature-based AC activation
- CO_2-based ventilation
- Water leak detection
- Humidity comfort control
- Smart window management
- Occupancy-full siren logic



Actions	Enabled	Name	Condition	Actions
Edit Remove	Yes	Air Quality	Match any: • co2-192.168.2.12 Level > 0.02 • temp-192.168.2.13 Temperature > 28.0 °C • humidity-192.168.2.14 Humidity > 65 %	Set fan-192.168.2.2 Status to High Set prayer-humidifier Status to false
Edit Remove	Yes	Air Quality (else)	Match all: • co2-192.168.2.12 Level <= 0.02 • temp-192.168.2.13 Temperature <= 28.0 °C • humidity-192.168.2.14 Humidity <= 65 %	Set fan-192.168.2.2 Status to Off
Edit Remove	Yes	Water Flow	water-level-monitor-192.168.3.11 Water Level >= 5.0 cm	Set siren-valve-192.168.3.3 On to true
Edit Remove	Yes	Water Flow (else)	water-level-monitor-192.168.3.11 Water Level <= 5.0 cm	Set siren-valve-192.168.3.3 On to false
Edit Remove	Yes	wudu-light-on-motion	wudu-motion-detector-192.168.3.12 On is true	Set wudu-light-192.168.3.1 Status to On
Edit Remove	Yes	prayer-light-on-motion	Match any: • prayer-motion-detector-192.168.2.11 On is true • prayer-motion-detector-2 On is true • prayer-motion-detector-3 On is true	Set prayer-light-192.168.2.4 Status to On
Edit Remove	Yes	prayer-light-on-motion (else)	Match all: • prayer-motion-detector-192.168.2.11 On is false • prayer-motion-detector-2 On is false • prayer-motion-detector-3 On is false	Set prayer-light-192.168.2.4 Status to Off
Edit Remove	Yes	wudu-light-on-motion (else)	wudu-motion-detector-192.168.3.12 On is false	Set wudu-light-192.168.3.1 Status to Off
Edit Remove	Yes	AC on High Temp	temp-192.168.2.13 Temperature > 32.0 °C	Set prayer-ac On to true
Edit Remove	Yes	AC on High Temp (else)	temp-192.168.2.13 Temperature <= 32.0 °C	Set prayer-ac On to false
Edit Remove	Yes	Close Windows on AC on	prayer-ac On is true	Set prayer-window On to false
Edit Remove	Yes	Turn on Siren on Full Mosque	Match all: • prayer-motion-detector-192.168.2.11 On is true • prayer-motion-detector-2 On is true • prayer-motion-detector-3 On is true	Set is-mosque-full-siren On to true
Edit Remove	Yes	Turn on Siren on Full Mosque (else)	Match any: • prayer-motion-detector-192.168.2.11 On is false • prayer-motion-detector-2 On is false • prayer-motion-detector-3 On is false	Set is-mosque-full-siren On to false
Edit Remove	Yes	Turn Humidifier on Low Humidity	humidity-192.168.2.14 Humidity < 30 %	Set prayer-humidifier Status to true
Edit Remove	Yes	open Window	Match all: • prayer-ac On is false • humidity-192.168.2.14 Humidity >= 60 %	Set prayer-window On to true

Figure 17: IoT server automation conditions

6. Dashboard and User Interfaces

The system includes two dashboard interfaces:

- Administrative dashboard
- User prayer-time dashboard

6.1 Administrative Dashboard

The admin dashboard allows:

- Real-time monitoring
- Device status visualization
- Alert observation
- Manual override control

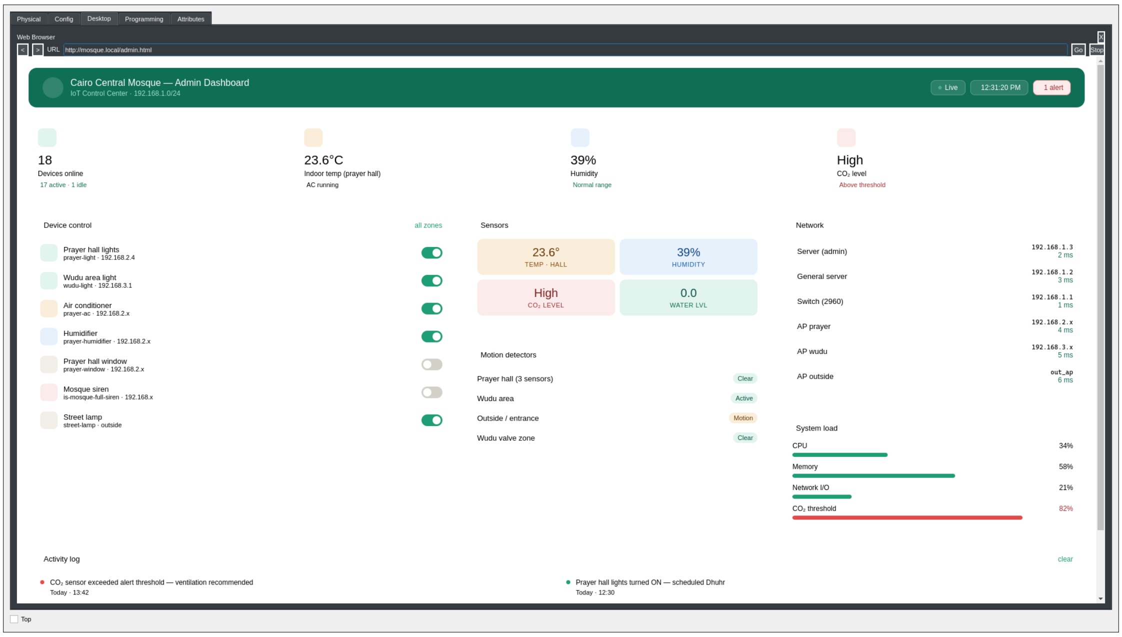


Figure 18: Administrative dashboard

6.2 Prayer-Time Dashboard

The user dashboard displays prayer schedules calculated using the USNO solar position algorithm.

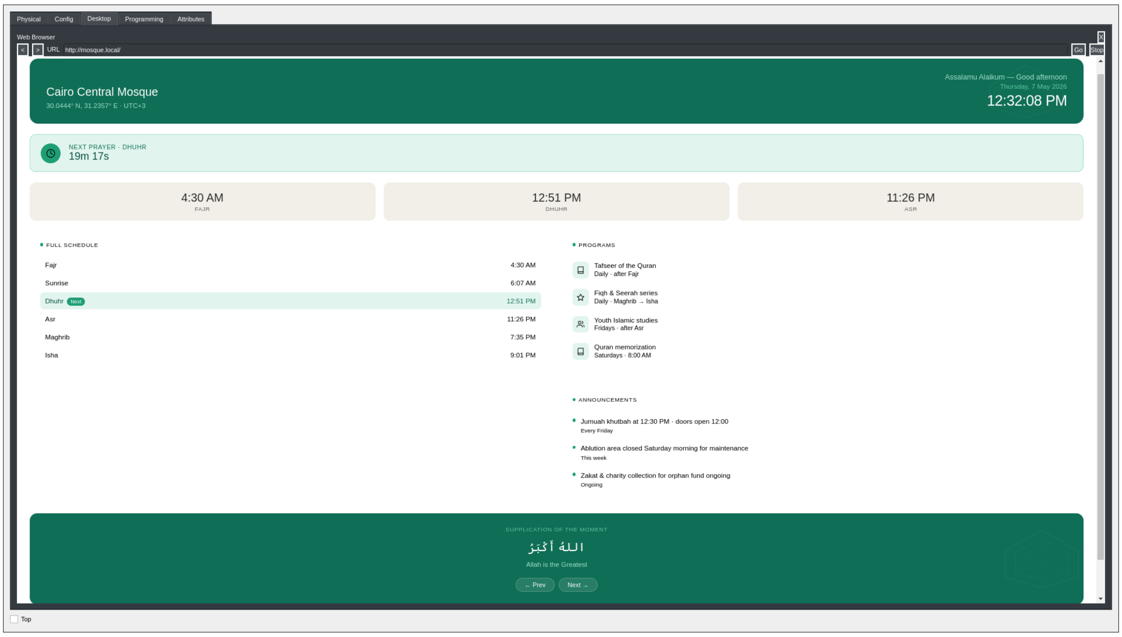


Figure 19: Prayer-time dashboard

7. System Evaluation

The simulation demonstrates successful implementation of:

- Energy-aware automation
- Occupancy monitoring
- Environmental comfort management

- Water conservation
- Real-time IoT monitoring

The modular design also allows future scalability through additional sensors and automation conditions.

The system remained responsive during simultaneous sensor activity and demonstrated reliable automated behavior during testing.

8. Challenges and Solutions

Several implementation challenges were encountered during development.

Challenge	**Solution**
No native solenoid valve in CPT	Used siren as simulation proxy
No external API support	Used USNO prayer-time calculations
Large number of devices	Implemented DHCP and DNS servers
Environmental testing difficulty	Used CPT environment controls
Server functionality limitations	Used separate infrastructure servers

Table 2: Implementation challenges and solutions

9. Future Improvements

Future development may include:

- Real ESP32 deployment
- MQTT communication
- Cloud monitoring
- Mobile application integration
- AI-based occupancy prediction
- Solar energy integration

10. Conclusion

The Smart Mosque Management System successfully demonstrates how IoT technologies can improve efficiency, comfort, and sustainability inside public religious facilities.

The implementation achieved:

- Automated environmental management
- Smart occupancy-aware control
- Water conservation
- Real-time monitoring

- Secure local-network communication

Despite Cisco Packet Tracer limitations, the simulation provides a realistic representation of a modular and scalable smart mosque ecosystem.

List of Figures